

RISHABH SINGH

Data Scientist, Forecasting, Analytics & Applied ML

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SUMMARY

IIT Madras graduate and Data Scientist with 2 years of commercial analytics experience, specializing in statistical modeling, time-series forecasting, mixed-effects modeling, experimentation, and end-to-end data product delivery. At Chryselys, co-engineered ForecastIQ, a live commercial pharma forecasting platform, and delivered Streamlit tools and automated dashboards adopted by pharma clients. Proficient in Python, SQL, R, Streamlit, scikit-learn, and cloud-based data pipelines. Adept at operationalizing models with engineering teams, translating analytical findings for non-technical stakeholders, and mentoring junior analysts across model validation and production-readiness workflows.

EXPERIENCE

Chryselys

Data Scientist (Forecasting, Analytics & Development)

Hyderabad, India
October 2024 – Present

- **ForecastIQ Product Delivery:** Architected and delivered core product components for [ForecastIQ](#), a live commercial pharma forecasting platform. Led dashboard development, applied product analytics, and evaluated the scenario planner module.
- **Model Performance and Operationalization:** Designed time-series cross-validation, error-tracking (MAPE, RMSE), and model comparison frameworks to support recurring performance reporting and model evaluation.
- **Model Deployment:** Collaborated with backend engineering teams to operationalize statistical models into production-ready pipelines.
- **Streamlit Tools and Enablement:** Engineered a Streamlit-based automated reporting and workflow tool that reduced manual preprocessing steps by approximately 40 percent, enabling non-technical stakeholders to independently explore segmentation outputs.
- **FFO Sizing and Mixed-Effects Models (Productization):** Strengthened and extended Mixed-Effects sizing models across 3 or more pharma client engagements. Hardened codebase and authored documentation to enable model reuse in go-to-market sizing exercises.
- **Marketing Mix Modeling (MMM):** Optimized feature engineering pipelines and developed promotion-response features to integrate MMM outputs with ML forecasting models for growth analytics.
- **Cross-Functional Delivery:** Partnered with Forecasting, Product, and Insights and Analytics teams to ship end-to-end forecasting solutions. Applied product analytics to align KPIs, translate findings into delivery priorities, and streamline reporting cadence between analytics and commercial teams.
- **Mentorship Cohort:** Coached 6 junior analysts on Python workflows, model validation standards, production-readiness checks, code reviews, and stakeholder-ready analytical presentations.

iNeuron.ai

Machine Learning Intern

Remote
March 2024 – April 2024

- Developed a phishing-domain detection model using Random Forests on approximately 147,000 URLs, applying feature engineering, model selection, and statistical validation to evaluate classification performance.
- Deployed the classifier as a Flask REST API on Microsoft Azure with a lightweight front-end for real-time URL classification by end-users.
- Authored reproducible evaluation notebooks and workflow documentation to support future extension and handover.

SKILLS

Languages: Python, SQL, R (basics), Bash, JavaScript (familiar)

ML & Statistics: Forecasting (ARIMA, ETS, Prophet, XGBoost), Mixed-Effects Models (LME/LMM), Marketing Mix Modeling (MMM), Regression Analysis, Tree-based Models (Random Forest, Gradient Boosting), SVM, Classification, Clustering, Segmentation, Time-Series Cross-Validation, Feature Engineering, Attribution Analytics, Statistical Modeling, Predictive Modeling, A/B Testing, Hypothesis Testing, Bayesian Inference, NLP, LLM Pipelines, Prompt Engineering, MCP, Tool Calling, Agentic Development (Claude, Codex), Growth Analytics, User Lifecycle Modeling, Growth Accounting, Product Analytics, Unsupervised ML, Entity Linkage, Graph Clustering, Fraud Detection

Tools & Frameworks: scikit-learn, pandas, NumPy, PySpark, PyTorch (familiar), TensorFlow (familiar), Streamlit, Flask, FastAPI, Jupyter Notebook, Git, GitHub, Tableau, Power BI (familiar), Figma, Dataiku, Databricks, Celery, Redis, Jira, Bitbucket, Dashboard Development, Automated Reporting Pipelines

Cloud & Infrastructure: Azure (App Service, Functions, Blob Storage), AWS (S3, Lambda, Bedrock, App Runner), PostgreSQL, MongoDB, Docker, Vercel, Render, REST APIs, CI/CD, MLOps, Data Pipelines, ETL, Feed-Based Recommendation Systems

EDUCATION

Indian Institute of Technology Madras (IIT Madras)

B.S. awarded May 2025

B.S. in Data Science and Applications, Minor: Economics and Finance, GPA: 7.89/10

Coursework: Machine Learning, Deep Learning, Neural Networks, Natural Language Processing (NLP), Probability & Statistics, Database Management Systems (DBMS), Software Engineering, Data Structures & Algorithms, Business Analytics, Econometrics, Game Theory, Operations Research

SELECTED PROJECTS

- **FraudNet**, Python, Fraud Detection, Entity Linkage, Fellegi-Sunter, Unsupervised ML, Graph Clustering, [github/fraud-network-viz](#)
Built unsupervised fraud detection on approximately 15,000 unlabeled signups from a signup-offer campaign. Applied Fellegi-Sunter weighted entity linkage across email aliases, device hash, card, and address to produce 251 suspected actor clusters. Shipped an interactive graph explorer ([live demo](#)) so reviewers can inspect why accounts were linked.
- **LLM YouTube Metadata Connector**, Python, MCP, Tool Calling, LLM Pipelines, YouTube API, Claude Desktop, [github/LLM-Youtube-Metadata-connector](#)
Built an MCP server that exposes YouTube Data API v3 tools to Claude Desktop for tool calling: fetch video data, generate SOP-compliant metadata, and publish updates through a two-step confirmation flow. Used to update 500+ YouTube videos. Implemented prompt engineering, deterministic validation, OAuth 2.0, and a human-in-the-loop review step so writes require explicit confirmation.
- **Parking Management System**, Flask, Vue.js, Redis, Celery, Azure, [parking.rishabhsingh.me](#)
Built a full-stack web application using Flask REST API and Vue.js. Leveraged Redis and Celery for asynchronous task queuing and deployed on Azure App Service with CI/CD. Implemented real-time slot booking, role-based access control, and an operational analytics dashboard.
- **Indian Housing ETL Pipeline**, PostgreSQL, MongoDB, Python, web scraping, [github/Indian-Housing-ETL-Pipeline](#)
Designed and implemented a scalable ETL pipeline covering scraping, cleaning, deduplication, and validation. Stored 180,000+ records across PostgreSQL and MongoDB for downstream statistical analysis and machine learning modeling.
- **Mixed-Effects Marketing Modeling**, Python, statsmodels, MMM, [Kaggle](#)
Developed hierarchical mixed-effects regression (LME) models to quantify marketing campaign elasticity. Decomposed global vs. channel-level effects to support data-driven budget allocation and ROI optimization using core Marketing Mix Modeling (MMM) techniques.
- **Audio Classification (UrbanSound8K)**, Python, TensorFlow, Librosa, Deep Learning, [Kaggle](#)
Built an end-to-end deep learning pipeline for urban audio classification. Engineered features using mel-spectrograms and MFCCs via Librosa and developed CNN models with TensorFlow, achieving strong accuracy on the UrbanSound8K benchmark dataset.
- **Compass (Bloglite)**, Flask, SQLAlchemy, REST API, deployment, [compass.rishabhsingh.me](#)
Developed a Flask-based multi-user blogging platform with SQLAlchemy ORM, JWT authentication, full CRUD operations, and RESTful API design. Containerized and deployed to cloud with environment-based configuration management.

CERTIFICATIONS & ACHIEVEMENTS

- **Microsoft Certified: Azure Fundamentals (AZ-900)**: cloud computing, Azure services, security, compliance, pricing
- **Microsoft Certified: Azure Data Fundamentals (DP-900)**: relational & non-relational data, Azure data services, analytics workloads
- **Microsoft Certified: Azure AI Fundamentals (AI-900)**: machine learning, computer vision, NLP, conversational AI on Azure
- **HarvardX: CS50 Introduction to Computer Science (2025)**: algorithms, data structures, C, Python, SQL, web development
- **HarvardX: R Basics (2025)**: statistical computing, data wrangling, visualization in R
- **UC Davis: Data Visualization with Tableau (2024)**: dashboards, visual analytics, storytelling with data
- **Rank 3**, IIT Madras CodeChef Programming Competition (2022): competitive programming, problem solving, algorithms
- Open source contributor to **CPython** ([python/cpython#134804](#)) and **conda-build** ([conda/conda-build#4782](#))